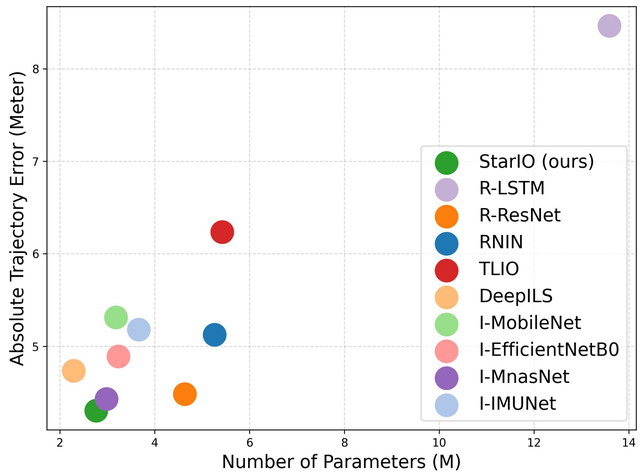
StarIO: A Lightweight Inertial Odometry for Nonlinear Motion

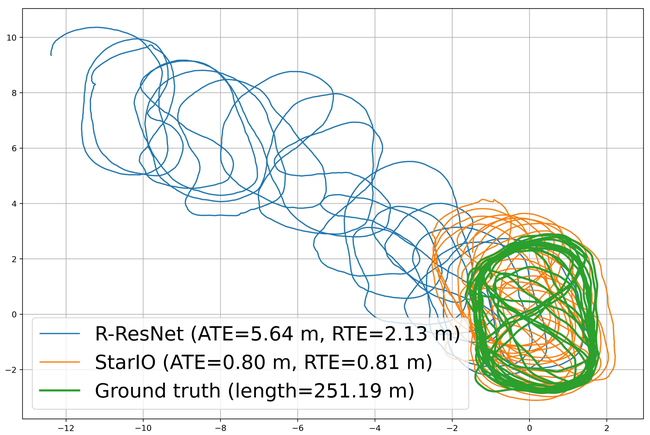
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Xuemin Hong, Ao Peng, Lingxiang Zheng, Yu Yang

Xiamen University

Motivations

- Inertial odometry (IO) serves as a promising localization solution for consumer-grade devices.
- Existing data-driven IO methods suffer from severe drift under complex nonlinear motion.
- They fail to effectively capture intrinsic nonlinear mappings between Inertial Measurement Unit (IMU) measurements and motion states.

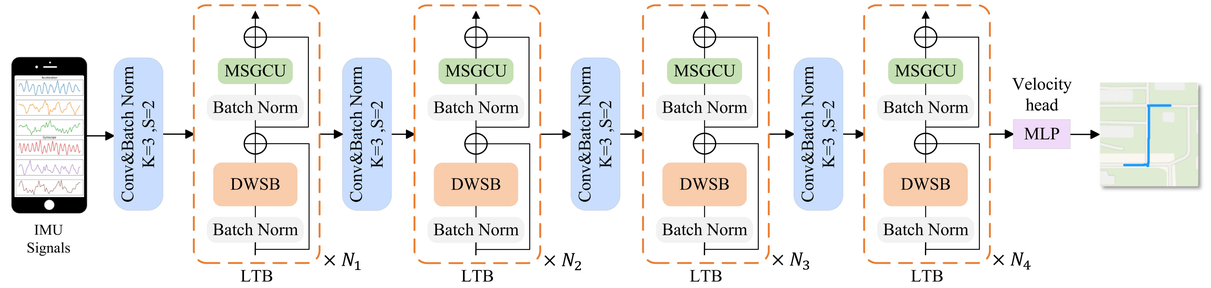


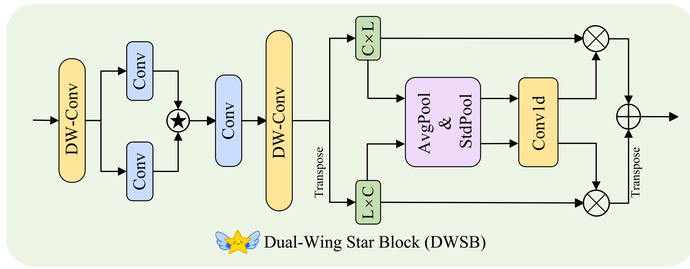


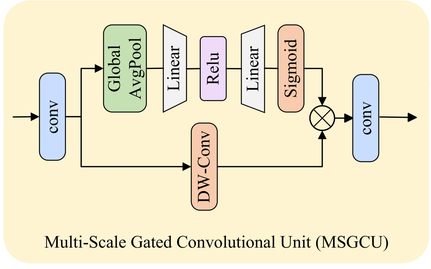
Representative trajectory from the OxIOD dataset

Methodology

- ✓ Project IMU signals into implicit high-dimensional nonlinear feature space via Star Operation to strengthen complex nonlinear motion modeling capability;
- ✓ Adopt a lightweight dual-wing attention mechanism to capture channel- and temporal-dimensional contextual motion information;
- ✓ Develop a multi-scale gated unit to aggregate local motion dynamics and contextual cues for integrated motion representation.









Experiment Result

TABLE I: Trajectory Prediction Accuracy of StarIO and Baseline Models. The best results are highlighted in bold.

Dataset	Metric	R-ResNet	R-LSTM	RNIN	I-IMUNet	I-MobileNet	I-MnasNet	I-EfficientNetB0	DeepILS	TLIO	StarIO	StarIO Improvement over R-ResNet
		Performance (meters)										
RoNIN	ATE	4.165	8.103	4.781	4.755	4.883	4.077	4.629	4.374	5.805	**3.948**	5.210%
	RTE	3.724	4.836	3.705	3.736	3.713	3.572	3.681	3.665	4.276	**3.554**	4.565%
RIDI	ATE	2.080	3.839	2.103	2.486	2.500	2.290	2.379	2.047	2.563	**2.033**	2.260%
	RTE	2.383	3.589	2.445	2.644	2.584	2.573	2.406	2.395	2.929	**2.274**	4.574%
RNIN	ATE	1.844	3.182	1.605	2.174	3.119	3.142	2.648	2.478	2.428	**1.455**	21.095%
	RTE	1.836	3.726	1.593	2.114	1.568	1.842	1.613	1.499	1.979	**1.370**	25.381%
IMUNet	ATE	6.428	8.108	6.445	9.130	6.846	8.593	6.479	6.918	10.240	**4.964**	22.775%
	RTE	4.802	6.381	4.463	6.404	4.809	5.609	4.458	5.638	5.735	**3.815**	20.554%
TLIO	ATE	1.333	2.145	1.300	1.354	1.369	1.373	1.724	1.258	1.832	**1.108**	16.879%
	RTE	3.705	3.685	3.627	4.060	0.842	**0.806**	0.959	0.818	5.162	3.145	15.115%
OxIOD	ATE	4.336	7.282	2.780	2.325	2.381	2.163	1.982	1.696	3.855	**1.484**	65.775%
	RTE	1.595	2.301	1.082	1.135	1.142	1.103	0.979	1.100	1.270	**0.856**	46.332%
Parameters (M)		4.635	13.589	5.259	3.659	3.179	2.979	3.232	**2.291**	5.424	2.762	40.410%
FLOPs (M)		38.252	57.473	72.032	18.838	34.559	25.256	28.831	**15.287**	39.436	25.122	34.325%
Peak Memory (MB)		22.835	70.755	25.490	38.274	67.257	**13.073**	25.631	17.746	60.026	20.458	10.409%

Inertial Odometry (IO): A lightweight localization scheme for pose and trajectory estimation using only inertial sensor data.
Inertial Measurement Unit (IMU): A low-cost inertial sensor that measures angular velocity and acceleration to characterize carrier motion.

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